

# NexaLink

Real-Time Communication Platform

## Full-Stack Product Blueprint & Implementation Plan

WebRTC   E2E Encrypted   AI-Powered   Privacy-First

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|               |                                            |
|---------------|--------------------------------------------|
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| Prepared By   | NexaLink Engineering Team                  |

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## 1. Executive Summary

NexaLink is a next-generation, browser-based real-time communication platform built on WebRTC, offering high-definition video calls, crystal-clear audio, persistent text messaging, and advanced collaboration tools — all wrapped in a privacy-first, end-to-end encrypted architecture. Unlike commodity solutions, NexaLink integrates an AI intelligence layer, stealth audio filtering, synthetic voice transmission, secure remote desktop control, and a suite of power-user features that set it apart from Zoom, Teams, and Google Meet.

**Mission:** Build the most feature-rich, privacy-respecting, and developer-friendly real-time communication platform that enterprises and power users can trust.

| Metric                             | Target                                        |
|------------------------------------|-----------------------------------------------|
| Max concurrent participants / room | 100                                           |
| Video latency (peer-to-peer)       | < 80 ms                                       |
| Audio latency                      | < 40 ms                                       |
| End-to-end encryption              | AES-256-GCM + DTLS-SRTP                       |
| Uptime SLA                         | 99.95%                                        |
| Mobile support                     | iOS 15+, Android 10+                          |
| Browser support                    | Chrome 90+, Firefox 88+, Safari 15+, Edge 90+ |

## 2. Platform Overview & Architecture

NexaLink follows a microservices architecture deployed on Kubernetes, with a WebRTC-native media plane, a RESTful + WebSocket control plane, and an AI sidecar layer. The system is designed for horizontal scalability, geographic distribution, and zero-knowledge privacy.

## High-Level Architecture Layers

| Layer                       | Components                                   | Tech                              |
|-----------------------------|----------------------------------------------|-----------------------------------|
| <b>Client (Browser/App)</b> | React SPA, WebRTC APIs, Canvas renderer      | React 18, TypeScript, Vite        |
| <b>Signalling Plane</b>     | Room management, SDP exchange, ICE brokering | Node.js, Socket.IO, Redis Pub/Sub |
| <b>Media Plane</b>          | SFU / MCU, codec negotiation, recording      | mediasoup / LiveKit, TURN/STUN    |
| <b>Application API</b>      | Auth, rooms, users, recordings, billing      | FastAPI (Python), PostgreSQL      |
| <b>AI Sidecar</b>           | Whisper ASR, TTS engine, meeting insights    | Python, Whisper, Coqui TTS        |
| <b>Privacy Engine</b>       | Alias management, metadata stripping         | Node.js, custom middleware        |
| <b>CDN / Storage</b>        | Recordings, avatars, file transfers          | S3-compatible, CloudFront         |
| <b>Observability</b>        | Metrics, tracing, alerting                   | Prometheus, Grafana, Jaeger       |

## 3. Complete Feature Set

NexaLink ships with a comprehensive set of real-time communication, collaboration, privacy, and AI-powered features across six categories.

### Video & Audio Core

- HD video (1080p) and ultra-HD (4K) with adaptive bitrate
- Spatial audio engine — 3D positional sound in multi-party calls
- Background blur, virtual backgrounds, and custom scenes
- AI-driven noise cancellation (echo, wind, keyboard, HVAC)
- Automatic gain control and dynamic range compression
- Hardware acceleration (H.264, VP9, AV1 codec support)
- Mirror / flip self-view; spotlight & pin participants

### Messaging & Collaboration

- Persistent in-room and direct message threads
- Rich text editor: markdown, code blocks, emoji reactions
- File sharing (drag & drop), inline image preview
- Threaded replies and message search
- Shared real-time whiteboard with shapes, sticky notes, arrows
- Live polls, Q&A queue, and hand-raise queue

- Breakout rooms — host splits and manages sub-groups
- Meeting agenda & timer widget pinned to screen

### Screen & Content Sharing

- Full-screen, window, or tab sharing with cursor indicator
- Annotation overlay on shared screen (all participants)
- Virtual pointer / laser pointer with coordinate tracking
- Picture-in-Picture (PiP) — keep call visible while sharing
- Multi-screen share — up to 4 sources simultaneously
- Remote scroll and zoom control on shared content

### Privacy & Alias Engine

- Privacy alias: replace real name, avatar, and email in-room
- Metadata stripping from all shared media before transmission
- Ephemeral rooms — zero persistence after call ends
- Selective recording consent — individual opt-in/out
- GDPR-compliant data handling with automated retention policy
- Zero-knowledge architecture — server cannot decrypt streams

### Stealth & Synthetic Voice

- High-gain whisper filter — transmit below ambient threshold
- Voice pitch and timbre morphing (privacy / accessibility)
- Text-to-speech engine with multiple voice profiles
- TTS interlock — type message and transmit as voice on demand
- Mute-with-AI: continue transcribing while muted (self-view only)
- Push-to-talk mode with configurable hotkey

### Remote Control & Security

- Companion desktop agent for secure remote input tunnelling
- Granular permission model: view-only, interact, full control
- Chaperone protocol: every remote action requires host confirmation
- Emergency kill-switch: instant revoke of all remote permissions
- Session recording with audit trail (who did what, when)
- Device trust verification before granting control
- Rate-limited input queue to prevent remote abuse

## Platform & UX Polish

- Responsive PWA — works on desktop, tablet, mobile browser
- Dark / light / high-contrast accessibility themes
- Keyboard-first navigation and full ARIA compliance (WCAG 2.1 AA)
- Custom branding / white-label mode for enterprise
- In-app diagnostics dashboard: bandwidth, packet loss, jitter
- Network quality indicator and auto-quality degradation
- Waiting room + lobby with host approval flow
- Calendar integration (Google, Outlook) for scheduled calls
- Webhook and REST API for third-party integrations

## 4. Phase 1 — Infrastructure & Media Pipeline

### Phase 1: Infrastructure & Media Pipeline Cloud, WebRTC, Privacy Alias Engine — Weeks 1–6

Establish the bedrock: production cloud infrastructure, a robust WebRTC media pipeline capable of handling hundreds of concurrent participants, and the foundational privacy-alias engine.

### Cloud Infrastructure

- Provision Kubernetes cluster (GKE / EKS) with auto-scaling node pools
- Set up CI/CD pipelines (GitHub Actions → ArgoCD) for zero-downtime deploys
- Configure private VPC, WAF, DDoS protection, and TLS termination
- Deploy Redis cluster for session management and pub/sub signalling
- Stand up PostgreSQL (primary + replica) with automated backups
- Configure object storage (S3/R2) for recordings and media

### WebRTC Media Pipeline

- Deploy mediasoup SFU with TURN/STUN (coturn) server fleet
- Implement SDP negotiation service and ICE candidate brokering
- Build room lifecycle manager: create, join, leave, destroy
- Enable adaptive bitrate control with network quality probing
- Implement DTLS-SRTP encryption for all media streams
- Add codec negotiation layer (VP8/VP9/H.264/AV1 prioritisation)
- Build recording pipeline: composite + individual track recording

### Privacy Alias Engine

- Design alias schema: ephemeral ID ↔ real user mapping (server-side only)

- Strip metadata from all inbound media (EXIF, device fingerprints)
- Build ephemeral room mode with post-session data wipe scheduler
- Implement selective recording consent flow and audit log

### Authentication & Identity

- OAuth 2.0 / OIDC integration (Google, GitHub, Microsoft, magic-link email)
- JWT-based session tokens with refresh rotation
- Role-based access control: owner, host, co-host, participant, viewer
- MFA (TOTP + WebAuthn) for enterprise accounts

## 5. Phase 2 — Stealth & Voice Intelligence

### Phase 2: Stealth & Voice Intelligence Whisper Filter, TTS, Audio AI — Weeks 7–11

Build the advanced audio intelligence stack including the high-gain whisper filter, AI noise cancellation, text-to-speech interlock, and voice morphing for privacy and accessibility.

### High-Gain Whisper Filter

- Implement WebAudio API gain node pipeline with real-time analysis
- Voice Activity Detection (VAD) using WebRTC built-in + custom ML model
- Whisper-mode: attenuate signal to sub-ambient level while preserving intelligibility
- User-configurable sensitivity threshold with visual feedback meter

### AI Noise Cancellation

- Integrate RNNoise / DeepFilterNet WASM module in browser audio chain
- Real-time suppression of: background music, keyboard clicks, HVAC, echo
- Fallback to browser echo-cancellation API when WASM unavailable
- Per-track noise profile: each participant's stream processed independently

### Text-to-Speech Engine

- Deploy Coqui TTS (open-source) with 10+ voice profiles
- TTS interlock UI: type → preview → transmit with single key press
- Voice personality library: gender, accent, speed, pitch controls
- Async TTS queue so host can queue multiple messages

### Voice Morphing & Privacy

- Real-time pitch shifting using WebAudio shift filter
- Timbre morphing profiles (no ML training required — DSP-based)
- Mute-with-transcription: local Whisper WASM continues listening

## 6. Phase 3 — Collaborative Interface

### Phase 3: Collaborative Interface Whiteboard, Screen Share, Virtual Pointer, PiP — Weeks 12–17

Deliver the rich collaboration surface: real-time whiteboard, advanced screen sharing with annotation, virtual pointer with coordinate broadcasting, and Picture-in-Picture so users never lose call context.

#### Screen Sharing Suite

- Implement `getDisplayMedia()` with tab/window/screen selector
- Build multi-source share: up to 4 streams via SFU simulcast
- Annotation overlay: SVG canvas layered above the remote video stream
- Annotation toolbar: pen, highlighter, arrow, shape, text, eraser
- Remote scroll control: host can delegate scroll permission to presenter

#### Virtual Pointer System

- Broadcast pointer ( $x\%$ ,  $y\%$ ) coordinates over signalling data channel
- Render laser-dot SVG overlay synced to shared screen dimensions
- Per-participant unique pointer colour and name label
- Pointer trail mode and spotlight mode (dim surroundings)

#### Picture-in-Picture Window

- Implement Document PiP API (Chrome 116+) with React portal fallback
- PiP shows: speaker grid, self-view, chat input
- Drag-and-dock: snap to corners; size presets (compact / expanded)
- Always-on-top mode for multi-monitor setups

#### Shared Whiteboard

- Canvas-based collaborative whiteboard using Fabric.js + CRDT sync
- Tools: pen, shapes, text, sticky notes, image upload, eraser
- Infinite canvas with zoom/pan; real-time multiplayer cursors
- Export whiteboard as PNG / PDF; import slide images as background
- Version history and undo/redo per participant

#### Meeting Engagement Tools

- Live polls: single/multi-choice, word cloud, rating scale
- Q&A queue with upvoting; host moderation dashboard
- Hand-raise queue with audio chime notification
- Breakout rooms: auto-assign or manual; countdown timer; broadcast to all rooms
- Meeting agenda widget: items, timers, presenter assignment



## 7. Phase 4 — Secure Remote Input & Control

### Phase 4: Secure Remote Input & Control    Desktop Agent, Input Tunnelling, Chaperone Protocol

Build the companion desktop agent, encrypted input tunnelling system, Chaperone confirmation protocol, and emergency kill-switch — giving hosts granular, audited control over remote interactions.

#### Companion Desktop Agent

- Electron-based agent for Windows / macOS / Linux
- Registers with server via mTLS certificate pinning
- System tray integration with quick status indicators
- Auto-update mechanism with signature verification
- Sandboxed input capture: keyboard, mouse (no file system access)

#### Input Tunnelling System

- Encrypted WebSocket channel (TLS 1.3) for input event stream
- Input serialisation: mouse delta, key code, modifier state
- Rate limiting: max 120 events/sec to prevent flooding
- Latency compensation: client-side prediction with server correction
- Separate data channel from media to avoid head-of-line blocking

#### Chaperone Protocol

- Every destructive action (file open, form submit) triggers host confirmation modal
- Confirmation timeout: auto-deny after 10 seconds if no response
- Configurable action allowlist: host pre-approves specific action types
- Full audit trail: timestamp, action type, approval/denial, actor
- Visual indicator on remote screen: coloured border when control is active

#### Emergency Kill-Switch

- Single-click (or hotkey **Ctrl+Shift+K**) revokes all remote permissions instantly
- Broadcasts revocation to all participants in room
- Post-revocation: 60-second cool-down before remote control can be re-granted
- Kill-switch event always logged and cannot be suppressed

## 8. Phase 5 — AI Layer & Smart Features

### Phase 5: AI Layer & Smart Features

#### Meeting Intelligence

- Action item extraction: detects task assignments and due dates

## 9. Phase 6 — Polish, Testing & Hardening

### Phase 6: Polish, Testing & Hardening    Load Testing, Security Audit, Performance, Launch — Week

Harden every layer of the platform, certify it under real load, complete the privacy guard audit, and ship a production-ready product.

#### Load & Stress Testing

- k6 load tests: simulate 500 concurrent participants per room
- Media stress test: 100 simultaneous HD streams through SFU
- Signalling storm test: 10,000 join/leave events per minute
- Database load test: 50k concurrent API requests; query optimisation
- Chaos engineering: random pod kill, network partition, disk full

#### Security Audit & Penetration Testing

- Third-party penetration test (OWASP Top 10 + WebRTC threat model)
- Static analysis (Semgrep, Snyk) integrated into CI
- Dependency vulnerability scanning with auto-PR remediation
- CSP, CORS, HSTS, and security headers hardening
- End-to-end encryption verification with independent key audit

#### Performance Optimisation

- Frontend: code splitting, lazy loading, service worker caching
- Core Web Vitals target: LCP < 1.5s, FID < 50ms, CLS < 0.05
- WebRTC: jitter buffer tuning, FEC, RTX for packet loss recovery
- API: query plan review, N+1 elimination, Redis caching layer

#### Accessibility & Compliance

- WCAG 2.1 AA full audit and remediation
- GDPR Article 30 data processing record; DPA templates
- SOC 2 Type I controls documentation
- CCPA opt-out flow and data export API

## Launch Readiness

- Beta program: 500 users across 3 enterprise customer pilots
- Runbook: incident response, escalation path, on-call rotation
- Status page (statuspage.io) with real-time component health
- Feature flags (LaunchDarkly) for gradual rollout
- Post-launch monitoring: 30-day war room with 15-min alert SLA

## 10. Technology Stack

| Layer                 | Technology                     | Rationale                               |
|-----------------------|--------------------------------|-----------------------------------------|
| <b>Frontend</b>       | React 18 + TypeScript + Vite   | Fast HMR, type safety, ecosystem        |
| <b>Styling</b>        | Tailwind CSS + Radix UI        | Utility-first, accessible primitives    |
| <b>State</b>          | Zustand + React Query          | Lightweight global state + server sync  |
| <b>WebRTC</b>         | mediasoup-client / LiveKit SDK | Production SFU with simulcast           |
| <b>Real-time</b>      | Socket.IO over WSS             | Fallback polling; room namespaces       |
| <b>Backend API</b>    | FastAPI (Python 3.12)          | Async, OpenAPI auto-docs, Pydantic      |
| <b>Auth</b>           | Auth.js + custom JWT           | OAuth, magic-link, MFA                  |
| <b>Database</b>       | PostgreSQL 16 + pgvector       | Vector search for transcript embeddings |
| <b>Cache / PubSub</b> | Redis 7 (Valkey)               | Session store, leaderboard, pub/sub     |
| <b>SFU</b>            | mediasoup 3 / LiveKit          | Low-latency, CPU-efficient SFU          |
| <b>TURN/STUN</b>      | coturn (self-hosted)           | ICE traversal for strict NAT            |
| <b>TTS</b>            | Coqui TTS (XTTS-v2)            | Open-source, voice cloning capable      |
| <b>Noise Cancel</b>   | DeepFilterNet WASM             | In-browser real-time suppression        |
| <b>Whiteboard</b>     | Fabric.js + Yjs CRDT           | Conflict-free collaborative canvas      |
| <b>Desktop Agent</b>  | Electron 30 + Node.js          | Cross-platform input capture            |
| <b>Search</b>         | Elasticsearch 8                | Full-text transcript search             |
| <b>Container</b>      | Docker + Kubernetes (K8s)      | Scalable orchestration                  |
| <b>CI/CD</b>          | GitHub Actions + ArgoCD        | GitOps continuous delivery              |
| <b>Monitoring</b>     | Prometheus + Grafana + Jaeger  | Metrics, dashboards, tracing            |
| <b>CDN</b>            | Cloudflare R2 + CloudFront     | Global media delivery                   |

## 11. Security & Privacy Architecture

## Transport Security

- All traffic over TLS 1.3 (no TLS 1.0/1.1); HSTS with 1-year max-age
- DTLS-SRTP for all WebRTC media streams — no plaintext media ever
- Certificate pinning in the desktop agent
- Strict CSP, X-Frame-Options: DENY, no inline scripts

## Data Encryption

- AES-256-GCM for all data at rest (recordings, transcripts, messages)
- Per-room ephemeral key pair for end-to-end encrypted rooms
- Key material never leaves the client in zero-knowledge mode
- PBKDF2 with 600k iterations for password-derived keys

## Access Control

- RBAC with least-privilege defaults: viewers cannot chat without host grant
- Rate limiting on all public endpoints (100 req/min per IP, 1000/min authenticated)
- CSRF double-submit cookie pattern on all state-mutating endpoints
- SQL injection prevention: ORM parameterised queries, no raw SQL

## Privacy Compliance

- GDPR: right to erasure implemented via cascading soft-delete + S3 lifecycle
- Data residency options: EU, US, APAC region selection at room creation
- Minimal collection: no third-party analytics SDK in the call room
- Privacy alias ensures server logs contain pseudonymous IDs only

## 12. Sprint Roadmap & Timeline

| Phase                               | Sprints           | Duration         | Key Deliverables                       |
|-------------------------------------|-------------------|------------------|----------------------------------------|
| <b>Phase 1: Infra &amp; Media</b>   | S1–S6             | Weeks 1–6        | K8s cluster, SFU, auth, privacy alias  |
| <b>Phase 2: Voice Intel</b>         | S7–S11            | Weeks 7–11       | Whisper filter, TTS, noise cancel      |
| <b>Phase 3: Collaboration</b>       | S12–S17           | Weeks 12–17      | Whiteboard, screen share, PiP, pointer |
| <b>Phase 4: Remote Control</b>      | S18–S23           | Weeks 18–23      | Desktop agent, Chaperone, kill-switch  |
| <b>Phase 5: AI Layer</b>            | S24–S29           | Weeks 24–29      | Transcription, summary, coach, search  |
| <b>Phase 6: Polish &amp; Launch</b> | S30–S36           | Weeks 30–36      | Load test, pentest, PWA, beta, GA      |
| <b>Total</b>                        | <b>36 Sprints</b> | <b>≈9 Months</b> | <b>Full production release</b>         |

## 13. Risk Register

| Risk                                    | Likelihood | Impact   | Mitigation                                |
|-----------------------------------------|------------|----------|-------------------------------------------|
| WebRTC browser incompatibility          | Medium     | High     | Fallback TURN relay + polyfill layer      |
| SFU scalability under load              | Medium     | Critical | Horizontal SFU scaling; load test early   |
| AI inference latency > 2s               | High       | Medium   | GPU autoscaling; cache common responses   |
| Desktop agent OS permissions denied     | Medium     | High     | Graceful degradation to view-only mode    |
| GDPR non-compliance in transcript store | Low        | Critical | Legal review; auto-purge scheduler        |
| Third-party TTS voice quality poor      | Medium     | Medium   | Maintain 3 TTS provider options           |
| Key engineer attrition                  | Medium     | High     | Pair programming; thorough documentation  |
| TURN server cost overrun                | Low        | Medium   | Peer-to-peer first; TURN only as fallback |

## 14. Success Metrics & KPIs

| KPI                              | Target      | Measurement Method         |
|----------------------------------|-------------|----------------------------|
| P2P call setup time              | < 3 seconds | Client telemetry           |
| Video frame drop rate            | < 1%        | RTC stats API              |
| Transcript accuracy (WER)        | < 8%        | Manual evaluation set      |
| Meeting summary (ROUGE-L)        | > 0.45      | NLP evaluation benchmark   |
| Remote input latency             | < 100 ms    | Input event timestamp diff |
| Monthly Active Users (MAU) — 6mo | 10,000      | Product analytics          |
| Net Promoter Score (NPS)         | > 45        | In-app survey (quarterly)  |
| Security incidents (Critical)    | 0           | SIEM alerting              |
| Uptime (rolling 30 days)         | > 99.95%    | External uptime monitor    |